

Light and shape in the work of Robert Fones

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Abstract

I describe some interesting visual phenomena in the work of the artist Robert Fones, including combinatorial use of simple forms to generate three-dimensional shapes, a preference for lighting from overhead and to the left, the convex–concave ambiguity, and subtle but effective cues to lighting direction.

Keywords

light, three-dimensional shape, color, visual arts

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Before there was vision science, there were visual arts. Our scientific understanding of vision has often advanced together with developments in the arts, as painters, architects, and sculptors have created techniques that use properties of human vision to guide our perception of objects and scenes. Our knowledge about phenomena such as linear perspective, lighting, shading, surface color, and the appearance of materials owes much to the attention of both scientists and artists (Cavanagh, 2005).

Robert Fones is a Toronto-based artist whose work is intriguing to viewers with an interest in visual perception. In one series of works, his ‘bevelled paintings’ create a sense of three-dimensional shape by combining a small set of shape-inducing features, such as Y- and Ψ -junctions, in a flexible, modular fashion (Figure 1; Fones, 2025; Getty, 2017). Some of these paintings depict scenes with partly abstracted but recognizable elements. Others depict logos and text, and disrupt readability by visually grouping parts of letters in unfamiliar ways. These works were painted from physical objects constructed by the artist, and give a vivid *trompe-l’oeil* impression of three-dimensional shape, particularly when viewed from a distance at which the surface texture of the paint and canvas is not visible.

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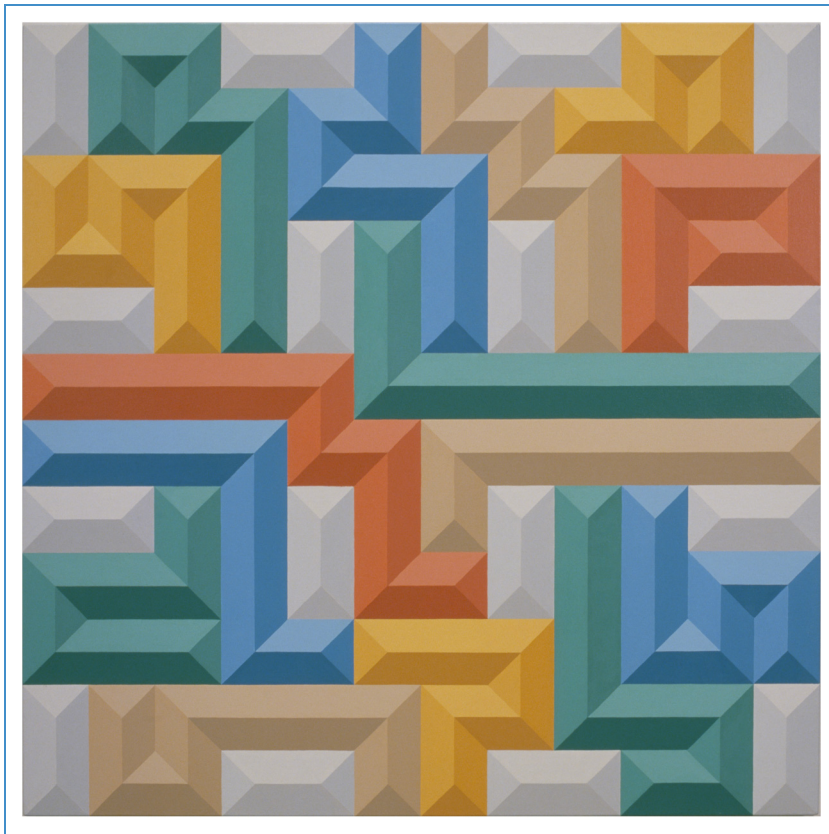


Figure 1. *Rebel Angels*. 2000. Acrylic on canvas. 101.6 × 101.6 cm. © Robert Fones.

Interestingly, the apparent lighting direction in the beveled paintings is consistently overhead and slightly to the left, as can be seen by comparing the brightness of elements in each colored region, e.g., elements that appear to face left are slightly brighter than elements of the same color that face right (Figure 1). Psychophysical work has shown this to be the most effective lighting direction for inducing percepts of three-dimensional shape, and it is common in a wide range of artwork from antiquity to modern times (Mamassian & Goutcher, 2001; Metzger, 1936/2006; Sun & Perona, 1998). Fones has also noted that he found color to be important for creating convincing three-dimensional shape in these paintings, and that achromatic elements were not as effective. This observation has echoes in psychophysical work as well (Kingdom, 2003).

In Figure 1, each colored strip appears to be convex. However, if the figure is turned upside down, many viewers perceive a change in shape, such that the colored strips appear concave, allowing the viewer to maintain a shape percept that is consistent with lighting from overhead (Ramachandran, 1988a). A perceptual bias toward convexity (Mamassian & Landy, 1998) can make this switch difficult to achieve, but looking away from the upside-down figure and back again helps to bring it about.

This reminds us that human vision relies on several assumptions, and that they may conflict with one another. When Figure 1 is upright, the light-from-overhead and convexity assumptions suggest the same three-dimensional shape, but when it is inverted, they suggest different shapes. When we look away from the inverted figure, we disrupt hysteresis effects that tend to make visual percepts stable over time (Leopold et al., 2002). When we look back again, we may perceive the colored

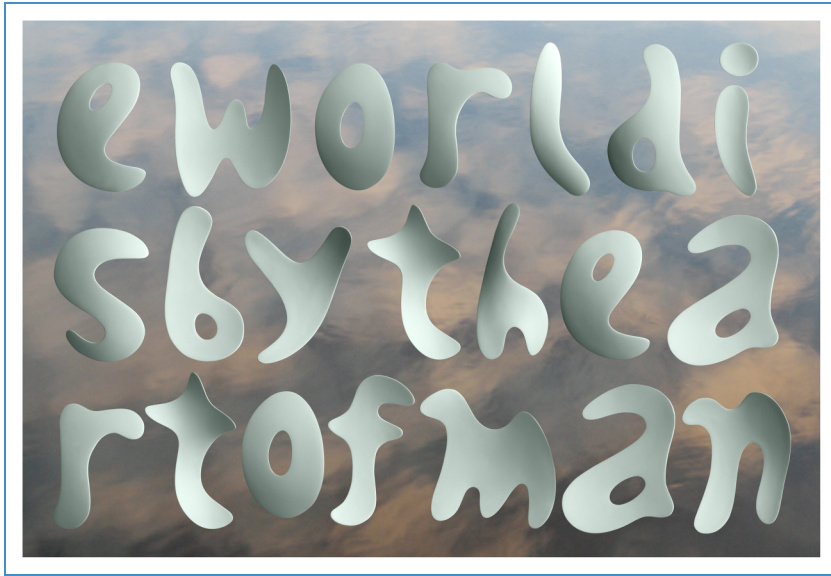


Figure 2. *Leviathan* #3. 2008. Color photograph on aluminum, edition of 3. 116.8 × 173.7 cm. © Robert Fones.

regions as convex or concave, depending on which of the implicit assumptions and visual cues that guide our percepts is stronger (Landy et al., 1995; Mamassian & Landy, 1998).

Also interesting are the *Leviathan* works, which depict lines from Hobbes's text (Figure 2). For these, Fones created thin, ceramic, convex or concave letters, and then photographed them and digitally merged the photographs into a composite image. Although the letters have been cropped out of the scenes where they were photographed, their smooth shading gradients still generate percepts of illuminated, three-dimensional shapes.

Surprisingly, most viewers find that turning Figure 2 upside down does *not* reverse the convexity or concavity of the letters. This may be because the letters are photographs of real objects and contain lighting cues that are not present in the artificial stimuli often used in perceptual experiments. For example, consider the letter 'e' at the top left of Figure 2. In addition to shading over its surface, the letter has a high-luminance boundary on some sides (mostly edges facing right) and a low-luminance boundary on others (edges facing left). These boundaries provide additional cues to the lighting direction, arising from the slight thickness of the object. As a result, when the figure is rotated by 180°, the lighting direction inferred by the viewer changes by approximately 180° as well (though see Jenkin et al., 2004), and the perceived three-dimensional shape of the letter remains the same. This explanation is speculative, and would be an interesting topic for empirical study: it suggests that luminance along a thin bounding line can have a decisive effect on the perceived shape of a much larger and otherwise ambiguous region. This would be consistent with the findings that perceived three-dimensional shape can be strongly influenced by the outlined shape of a region (Ramachandran, 1988a), and by nearby cues to lighting direction (Ramachandran, 1988b; Morgenstern et al., 2011).

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Ethical Approval

This manuscript presents an informal discussion of artwork. No data was collected.

Author Contribution(s)

Richard F. Murray: Conceptualization; Investigation; Writing – original draft; Writing – review & editing.

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Data Availability Statement

No data was collected for this paper.

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